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Opinion

A practical guide to spatial transcriptomics: lessons from over 1000 samples

Daniela Grases[1] and Eduard Porta-Pardo[1,2,*]

Spatial transcriptomics (ST) enables the *in situ* mapping of gene expression, revolutionizing our ability to study tissue organization and cellular interactions. However, many groups struggle with practical barriers to implementation, including platform selection, sample quality, and experimental scalability. We provide a practical guide to ST, informed by the processing and analysis of over 1000 spatial samples across multiple ST platforms. We outline best practices for experimental design, tissue handling, sequencing, and computational analysis, with special attention to clinical samples. Our goal is to translate hands-on experience into recommendations that support robust, reproducible spatial workflows. This guide is designed to assist researchers at all levels: from those designing their first spatial experiment to groups aiming to integrate ST into large-scale studies.

Highlights

Spatial transcriptomics (ST) has matured into a multidisciplinary effort, where tight coordination between molecular biologists, pathologists, histotechnologists, and computational analysts is now recognized as critical to success.

RNA quality metrics like DV200 and RNA integrity number (RIN) still guide expectations, but recent work shows that even below-threshold samples can yield biologically meaningful data.

Sequencing depth guidelines are evolving: formalin-fixed paraffin-embedded (FFPE) Visium experiments often benefit from 100–120 k reads/spot, well above the long-standing 25 k standard.

For targeted imaging platforms such as Xenium, larger panels may reduce per-gene sensitivity, highlighting the trade-off between breadth and depth of detection.

Preventing batch effects through randomization, replication, and comprehensive metadata tracking remains the most reliable strategy, as computational corrections can go only so far.

From concept to tissue: practical considerations in ST

ST refers to a set of technologies that allow researchers to measure gene expression directly within tissue sections, preserving the spatial location of each measurement. Unlike conventional RNA sequencing, which analyzes homogenized or dissociated samples, ST maintains the native architecture of the tissue, enabling the study of cellular neighborhoods, tissue organization, and microenvironmental gradients. This added spatial context has proven critical for understanding development, disease, and therapeutic response [1–10].

As adoption grows across biomedical research, the emphasis is increasingly shifting from proof-of-concept studies to robust, scalable experimental pipelines. Yet despite the proliferation of new platforms and protocols, many groups face persistent challenges when designing and executing spatial experiments, particularly in tissue handling, platform selection, and wet-lab execution. These early-stage bottlenecks often determine the success or failure of downstream analyses, but they are rarely addressed in depth in existing reviews, which tend to focus on computational workflows or benchmarking [11–23].

This opinion article aims to fill that gap. Drawing on experience from over 1000 spatial samples profiled across Visium, Visium HD, and Xenium platforms (Figure 1A,B), we offer a practical guide to spatial experimental design, centered on real-world decisions that often go underdiscussed. We focus on how to choose the right platform for a biological question, how to optimize tissue handling and sequencing depth, and how to scale experiments from pilot studies to translational pipelines involving dozens or hundreds of samples. These foundational choices are often underestimated, yet they are the basis of reproducible, interpretable, and ultimately impactful ST studies.

Step 1. Defining the research question

The most important decision in any ST experiment comes before the tissue is sectioned: is spatial resolution essential for answering your biological question?

[1]Josep Carreras Leukaemia Research Institute, Badalona, Spain
[2]Barcelona Supercomputing Center (BSC), Barcelona, Spain

*Correspondence:
eporta@carrerasresearch.org
(E. Porta-Pardo).

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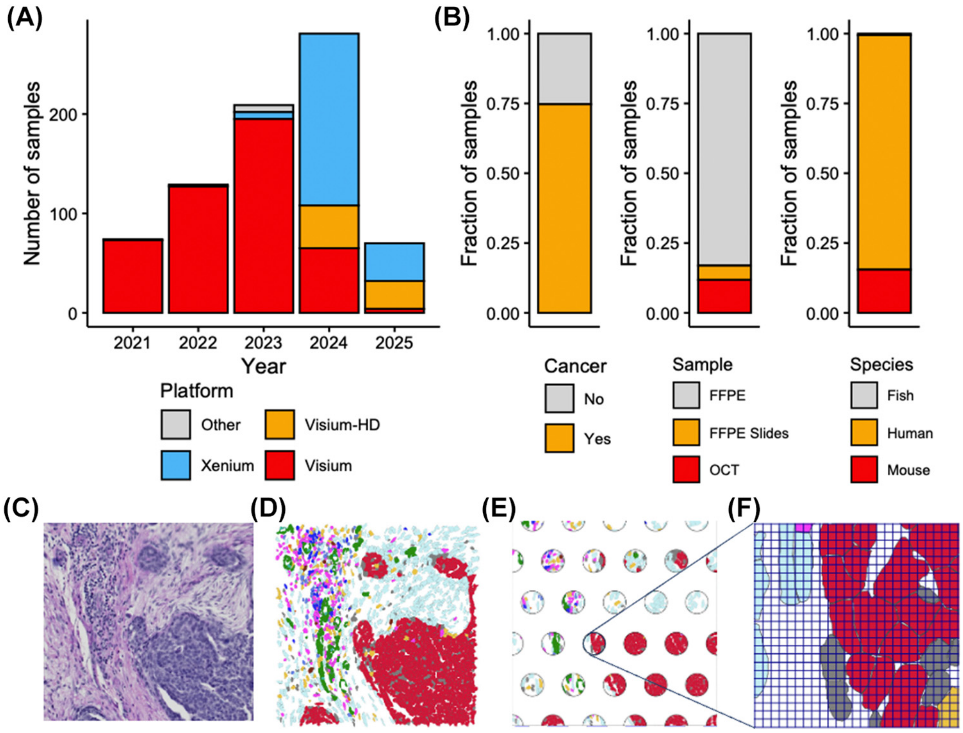

Figure 1. Our experience with spatial transcriptomics (ST) and differences in resolution between platforms. (A) Number of samples processed at our laboratory over the last 5 years, stratified by ST platform. Not all processed samples are shown due to confidentiality constraints. (B) Fraction of samples according to whether they are cancer-related or not (left), type of embedding (middle), and species (right). (C) Hematoxylin and eosin (H&E) image of an ovarian tumor. (D) The same tumor area with cells segmented and colored according to the cell type annotated from Xenium data. (E) Overlay of Visium spots (55 μm) in the same area, illustrating the oligo-cell resolution of this platform. (F) Zoom into one of the spots from 'E' with an overlay of the Visium HD grid (2 μm), showing the subcellular resolution of this platform. Abbreviations: FFPE, formalin-fixed paraffin-embedded; OCT, optimal cutting temperature compound.

ST excels when the goal is to understand how cell–cell interactions, tissue architecture, or micro-environmental gradients shape biological processes [24–27]. But questions focused on global transcriptional differences, such as bulk comparisons across timepoints or treatment arms, conventional RNA-seq or single-cell RNA-seq may be more appropriate [28].

Key considerations include the expected spatial heterogeneity of your tissue, the anatomical features of interest, and the scale at which biological patterns are likely to emerge. Do you need subcellular detail to study spatially restricted programs [29]?

This, in turn, determines platform resolution requirements. Traditional spot-based methods like Visium (~55 μm) suit domain-level analysis. However, newer platforms such as Visium HD achieve single-cell or even subcellular resolution while preserving whole-transcriptome coverage. Imaging-based platforms like Xenium, CosMx, or MERFISH also offer single-cell to subcellular resolution, typically using targeted panels, and are particularly effective for profiling fine-scale spatial features or validating known biomarkers.

Another essential variable is transcriptome breadth. Whole-transcriptome profiling (~18 000 genes) supports discovery-driven research, while targeted panels (~500–5000 genes) work well for hypothesis-driven studies or clinical applications. Panel size, resolution, and RNA quality must be considered together.

In short, designing a successful ST experiment means mapping your biological question onto three axes: spatial resolution, gene coverage, and tissue characteristics. Aligning these with platform capabilities prevents overengineering and ensures meaningful insights.

Step 2. Assemble the right team

ST is one of the most multidisciplinary workflows in modern molecular biology. Success hinges not just on technology or budget, but on assembling the right team, and involving them early. At a minimum, spatial projects require coordinated input from three domains: wet lab, pathology, and bioinformatics. Additionally, researchers new to ST are strongly encouraged to consult experienced labs or core facilities early in the planning process to avoid common pitfalls and platform-specific challenges.

The laboratory technician executes technically demanding protocols with minimal room for error. RNA integrity, hybridization efficiency, and spatial signal all depend on precise sectioning, reagent handling, and temperature control. Even small missteps (incorrect blade angle, slide detachment, uneven staining) can derail an entire experiment. Ideally, the same technician should process all samples within a study to minimize variation.

Pathologists provide critical insight into tissue quality, morphology, and region-of-interest (ROI) selection. Their input ensures that spatial data reflect biological architecture rather than technical artifacts. This is especially vital in clinical samples, where variability is high and tissue availability is limited. In multi-sample studies, pathologists also standardize ROI selection across samples for comparability.

Bioinformaticians translate raw spatial data into interpretable biology. They handle platform-specific preprocessing, normalization, and spatial analysis, and often lead integration with scRNA-seq, proteomics, or clinical metadata. Their input is essential even before data generation, particularly in platform selection and experimental power calculations.

Most failed spatial experiments are not due to a single mistake, but a lack of coordination between these roles. Effective communication between domains prevents wasted samples, poor data quality, and misaligned interpretations. Whether in academic labs or clinical consortia, assigning a project lead who bridges technical and biological perspectives is one of the simplest, most effective ways to ensure spatial success.

Step 3. Experimental design, controls, and statistical power

ST experiments are shaped by a unique set of design challenges: limited tissue, high cost per sample, and the need to resolve spatial heterogeneity across biological and technical dimensions.

A critical early decision is how many biological replicates and ROIs are needed to detect meaningful patterns (see Box S1 in the supplemental information online). Unlike single-cell RNA-seq, spatial data is highly sensitive to ROI selection, tissue orientation, and section quality [30]. Underpowered studies, often with too few samples or too narrow spatial coverage, risk missing key biological signals or over-interpreting noise. We recommend pilot studies and, when possible, data-driven power calculations to estimate required sample sizes [31–34]. When that is not possible, we guide users by simulating power curves based on expected effect sizes and tissue variability. This is often enough to balance the trade-offs between number of regions per sample and number of biological replicates based on budget.

Batch effects are another common pitfall. Differences in sectioning, slide placement, staining, imaging, or sequencing depth can obscure true biological variation. While computational tools can help correct some of these effects, prevention is far more effective. Key strategies include randomizing sample layout, processing all samples under consistent conditions, and systematically tracking metadata (see Note S1 in the supplemental information online).

The choice between full tissue sections and tissue microarrays (TMAs) depends on study goals. TMAs allow efficient profiling of many samples on a single slide (ideal for screening or clinical cohorts) but provide limited spatial context per sample [30]. Full sections offer rich architectural information but are more costly to scale.

Finally, whenever possible, include positive and negative controls to monitor technical variability, validate assay performance, and anchor cross-sample comparisons. These can include reference tissues, known marker regions, or replicate sections from the same block. While it is not necessary to include control tissue on every slide, doing so is recommended when space allows, particularly in imaging-based platforms with large capture areas, pilot studies, or when testing new panels or protocols.

In sum, thoughtful experimental design is essential to extract meaningful insights from spatial data. Budget and sample constraints are real, but careful planning, smart randomization, and quality control can dramatically improve robustness, reproducibility, and biological interpretability.

Step 4. Tissue selection, processing, and quality control

Tissue quality is one of the most overlooked determinants of ST success. From preservation method to sectioning conditions, small pre-analytical decisions can have large downstream effects on data quality and interpretability (Box S2 in the supplemental information online).

Preservation strategy is often dictated by study context. Fresh-frozen (FF) tissue generally provides higher RNA integrity and enables full-transcriptome analysis, but requires rapid freezing, careful cryosectioning, and strict cold chain maintenance. Formalin-fixed paraffin-embedded (FFPE) tissue offers better morphological preservation and is more compatible with clinical workflows, but often yields fragmented RNA. Many spatial platforms now support FFPE, but degraded input remains a major challenge.

When preparing FFPE blocks additional care during embedding can improve both RNA quality and section integrity. We recommend minimizing the delay between fixation and embedding, using fresh xylene and paraffin, and ensuring the final paraffin bath is free of residual xylene. Although researchers often work with blocks prepared externally (e.g., in hospital pathology labs).

RNA integrity must be assessed before proceeding. For FF samples, RNA integrity number (RIN) ≥7 is preferred, though newer protocols may tolerate RIN ≥4. For FFPE, DV200 (the percentage of RNA fragments >200 nt) is more relevant. Old Visium protocols recommended ≥50% DV200, but current versions of the platform (Visium V2, Visium HD) can work with ≥30%. Xenium and CosMx tolerate lower values depending on the panel.

That said, these thresholds are not absolute. In our experience, DV200 does not always correlate linearly with spatial data quality. For Visium, samples below 30% DV200 may still yield biologically meaningful results, albeit with reduced genes per spot (Figure 2A). In contrast, Xenium's performance is more tightly coupled to DV200, particularly when using the higher-plex Xenium Prime panel (Figure 2B). Median transcript counts drop more sharply in degraded samples with Prime

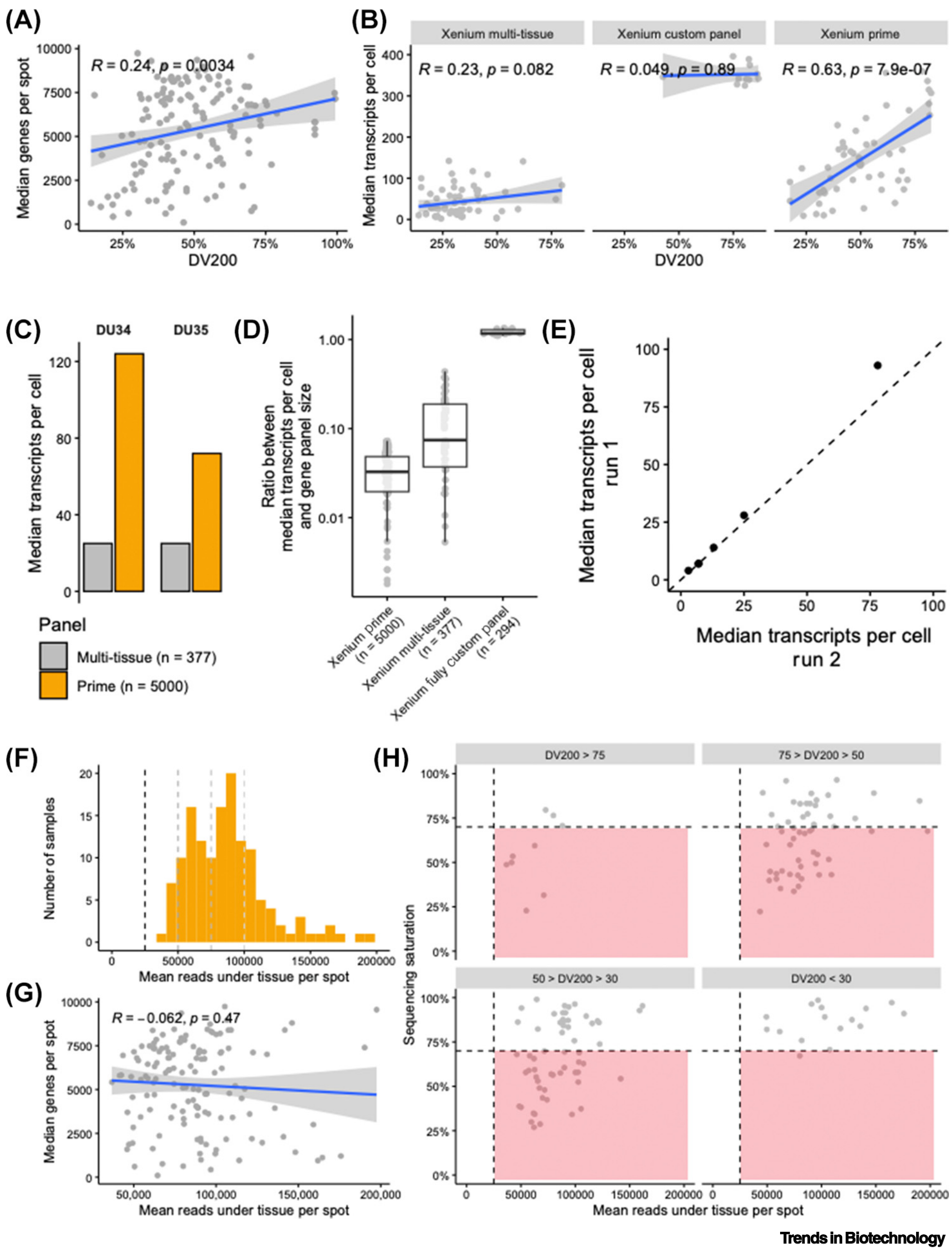

Figure 2. Data quality and sequencing considerations for Visium and Xenium. (A) Correlation between DV200 (the percentage of RNA fragments >200 nt) of the FFPE block (*x*-axis) and the median genes per spot obtained in Visium (*y*-axis) across 137 tumor samples. Higher DV200 values, reflecting better RNA integrity, generally yield more detected genes per spot. (B) Correlation between DV200 of the FFPE block (*x*-axis) and the median transcripts per cell (*y*-axis) obtained with Xenium, shown separately for different gene panels. Larger panels tend to recover more transcripts, but RNA quality still plays a role. (C) Median transcripts per cell (*y*-axis) obtained in two bladder cancer samples analyzed with the Xenium multi-tissue (gray) and Xenium Prime (orange) panels, illustrating that larger panels increase detection sensitivity. (D) Ratio between median transcripts per cell and panel size (*y*-axis) depending on the type of Xenium panel (*x*-axis), showing that smaller panels can yield more transcripts per gene due to probe allocation efficiency. (E) Correlation in median transcripts per cell between technical replicates using the Xenium multi-tissue panel, demonstrating high reproducibility. (F) Distribution of the mean reads under tissue per spot for our collection of Visium cancer samples (n = 137). The vertical dashed line marks the 10x Genomics recommendation of ≥25

(Figure legend continued at the bottom of the next page.)

than with the lower-plex Multi-Tissue panel. Therefore, when working with FFPE blocks near the lower DV200 limit smaller targeted panels, such as Xenium MultiTissue, tend to perform better than high-plex panels, as they focus on robust, highly expressed transcripts and are generally more tolerant of RNA degradation.

Sectioning requires precision and consistency. We recommend 10 μm for fresh-frozen and 5 μm for FFPE samples. All sectioning should be done in RNase-free conditions, with clean blades and minimal tissue handling. Always discard the first one or two sections to remove oxidized surface material. After cutting, FFPE sections should be dried at 42°C for 3 h (overnight, in the case of TMAs) and then stored in a sealed desiccator at room temperature for up to 6 months. Fresh-frozen sections should be stored at –80°C immediately and used within 4 weeks to minimize RNA degradation.

Difficult tissues require additional care. Fat-rich samples (e.g., breast, liver) benefit from colder cryostat temperatures (–30°C) and slightly thicker sections (10–12 μm) in optimal cutting temperature compound (OCT). For tissues composed almost entirely of adipose content, where RNA yield is extremely low and morphology is particularly fragile, one can use thicker sections, up to 25 μm. In FFPE, lowering the water bath temperature (~38°C) helps prevent disintegration. Calcified tissues require EDTA-based decalcification and careful embedding to preserve both structure and RNA [35,36].

Visual QC is equally essential. H&E or DAPI-stained sections provide critical insight into nuclear morphology, necrosis, and tissue integrity. For imaging-based platforms, intact nuclei and minimal tissue damage are strong predictors of success, even in samples with borderline DV200.

Ultimately, high-quality spatial data relies not only on RNA metrics but on consistent, context-aware tissue handling across every step of the workflow.

Step 5. Spatial platform selection

Choosing the right ST platform is a critical design decision that must align with your biological question, tissue constraints, and downstream goals [37]. The main trade-offs involve three interdependent axes: spatial resolution, gene coverage, and input quality (Figure 1C–F). Importantly, while our guide focuses on 10x Genomics platforms due to sample volume and reproducibility, alternative platforms (e.g., MERSCOPE, seqFISH, Slide-seq) may be better suited in certain settings, especially when driven by local expertise or existing laboratory infrastructure.

Whole-transcriptome platforms such as Visium and Visium HD offer unbiased detection of ~18 000 genes, making them ideal for exploratory or atlas-scale studies. Visium provides moderate resolution (~55 μm), capturing multicellular domains, while Visium HD achieves subcellular resolution (~2 μm) while having whole-transcriptome breadth. These methods are best suited for fresh-frozen or high-quality FFPE samples. While 10x Genomics provides protocol-specific minimum sequencing depth recommendations, we routinely sequence Visium FFPE samples at 100 000–120 000 reads per spot, and Visium HD samples even deeper, to ensure sufficient coverage of low-abundance transcripts and robust downstream analysis. These values reflect our empirical experience and tend to exceed official guidelines, particularly for clinical or degraded samples.

000 reads per spot; many samples exceed this. (G) Relationship between the mean reads per spot (*x*-axis) and the median genes per spot (*y*-axis) for Visium data, showing no correlation beyond the recommended depth, indicating that higher sequencing does not necessarily increase gene detection. (H) Relationship between sequencing saturation (*y*-axis) and the number of reads per spot (*x*-axis), stratified by DV200 of the FFPE block. The vertical dashed line marks the 25 000-reads-per-spot minimum, and the horizontal dashed line marks 70% saturation. Red rectangles indicate under-sequenced samples. Based on these results, we recommend >100 000 reads per spot for Visium libraries from samples with DV200 above 30% to ensure robust detection. Abbreviation: FFPE, formalin-fixed paraffin-embedded.

In contrast, imaging-based platforms like Xenium, CosMx, and MERFISH rely on pre-selected gene panels to achieve high spatial resolution at the single-cell or subcellular level. These approaches are well suited to hypothesis-driven experiments or biomarker validation, especially in clinical samples with limited RNA quality. While their targeted nature limits discovery potential, they offer significant practical advantages in tissue flexibility, processing scalability, and spatial precision.

One major differentiator is tissue area coverage. Visium's fixed capture area (~6.5 × 6.5 mm or 11 mm × 11 mm) restricts its use for full sections or tissue microarrays (TMAs). In contrast, Xenium and CosMx can image large contiguous areas across multiple slides per run, enabling efficient profiling of dozens of TMA cores or full-slide clinical specimens. This makes them particularly well suited for high-throughput or pathology-integrated pipelines.

Panel size also influences sensitivity. As shown in Figure 2D, expanding the number of genes in imaging-based platforms does not linearly increase transcript detection. For example, Xenium Prime's 5000-gene panel yields broader coverage but lower per-gene sensitivity than the Multi-Tissue panel, especially in FFPE samples with modest RNA quality. When using large panels in degraded tissue, the lower signal-to-noise ratio can affect both cell typing and spatial feature detection. That being said, Xenium measurements also show high technical reproducibility, with strong correlations in median transcripts per cell between replicate experiments (Figure 2E), highlighting the platform's robustness when sample preparation and imaging conditions are consistent.

RNA quality remains a key constraint. Visium and Visium HD require DV200 ≥30% for reliable FFPE performance. By contrast, Xenium and CosMx can tolerate DV200 values as low as 10%, though degraded samples may still affect panel performance and spatial resolution. Platform-specific sensitivity to RNA fragmentation must be considered when selecting both the technology and the gene panel.

Other considerations include workflow complexity and scalability. Sequencing-based methods require additional library prep and bioinformatics steps that may limit throughput. Imaging platforms bypass this by directly generating spatially resolved transcript data from the slide, simplifying integration with pathology laboratories and clinical workflows.

Cost considerations also influence platform choice, but absolute figures vary between countries for multiple reasons (e.g., currency exchange rates, third-party distributors, temporary discounts). While exact prices change over time, a useful reference is the relative cost per square millimeter of capture area. Using a Visium v2 CytAssist FFPE run as the baseline, Visium HD costs ~75% more, Xenium with a small panel ~30% less, and Xenium Prime is almost equivalent (~3% more). These values include platform reagents and sequencing (for Visium) or panel reagents (for Xenium), but exclude labor, instrument depreciation/maintenance, and ancillary consumables.

In summary, platform choice should reflect the study's resolution requirements, gene discovery needs, tissue quality, and scalability goals. A summary of trade-offs across platforms is provided in Table 1.

Step 6. Execution of the experiment

Once tissue quality has been confirmed and the platform selected, successful execution depends on disciplined laboratory practices. ST protocols are often rigid and unforgiving, especially in the wet-lab stages, so small procedural errors can compromise entire experiments.

Table 1. Overview of commonly used ST platforms

Platform (type)	Resolution and panel type	Sample types	RNA quality	Best use cases
Visium (FF, sequencing)	~55 μm; whole transcriptome	Human, mouse, all species (polyA+)	RIN ≥ 7 (≥4 w/ CytAssist)	Broad discovery in fresh tissue
Visium (FFPE, sequencing)	~55 μm; whole transcriptome	Human, mouse FFPE	DV200 ≥50% (≥30% w/ CytAssist)	Archived samples; full profiling
Visium HD (sequencing)	~2 μm; whole transcriptome	Human, mouse FFPE or OCT	RIN ≥4; DV200 ≥30%	High-res + whole transcriptome
Xenium (imaging)	Subcellular; targeted (up to 5000 genes; customizable)	Human, mouse FFPE or fresh-frozen; non-model (custom)	DV200 ≥10%	Cell typing; high-res profiling; cross-species (custom panels)
CosMx (imaging)	Subcellular; targeted panels or whole-transcriptome	Human, mouse FFPE or fresh-frozen	DV200 ≥10%	Multiplexed profiling; spatial cell state mapping
MERFISH/ seqFISH (imaging)	Subcellular; highly multiplexed (customizable)	Fresh-frozen; FFPE (with optimization)	Protocol-dependent	Deep profiling in microscopy-capable labs
Stereo-seq (sequencing)	500 nm; whole transcriptome (species-specific probes)	Human, mouse, non-model (custom probes)	RIN ≥7 recommended	Nanoscale mapping; large area profiling
Non-model species	Varies by platform and probe design	Visium (polyA+), Xenium, Stereo-seq	Variable	Cross-species studies (requires custom panels or transcriptomes)

Pre-experiment setup is essential (see Note S2 in the supplemental information online). Reagents must be fresh, properly stored, and equilibrated to the correct temperature. Alcohols, enzymes, and buffer solutions should be handled precisely; pipetting accuracy and timing can have an outsized effect, particularly in hybridization and washing steps. Lot numbers and expiration dates should be recorded.

Sectioning and sample placement remain among the most failure-prone stages. In FFPE blocks, tissue visibility is usually sufficient to locate regions of interest (ROIs). In OCT-embedded blocks, however, pale tissue and poor contrast make orientation difficult. When morphology is ambiguous, we recommend staining a test section or referring to an adjacent H&E-stained slide before collecting the experimental section.

If the ROI occupies only a small portion of the block, outline it with a scalpel at the surface, or re-embed the scored area into a new block. This approach also supports the creation of custom TMAs, which are ideal for parallel analysis across many samples under uniform conditions. Alternatively, if the ROI is clearly visible in the section, one can isolate it at the section level. This can be done by floating the section on a cold-water bath, using a clean razor blade to excise the ROI, and then transferring only the desired piece again to the warm bath for final mounting, preserving the rest of the tissue and avoiding block scoring. Sectioning might need to be adapted for certain tissues that are particularly challenging (see Box S3 in the supplemental information online).

Working with TMAs introduces unique challenges. Core detachment during sectioning is common, particularly for small or central cores that lack sufficient paraffin support. Core dropout can affect 15–30% of samples per slide. Strategies to reduce this include using larger cores (1.5–2.0 mm), trimming excess paraffin before sectioning, and pre-preparing multiple serial sections in case of dropout or damage.

Tissue alignment is another critical step, especially in protocols like Visium CytAssist. This bridge instrument enables ST on standard glass slides by transferring RNA from the tissue section into a separate barcoded capture slide via direct alignment and hybridization. This decouples tissue sectioning and histological preparation from the capture step, allowing samples to be prepared on standard slides and later processed in a centralized facility. This logistical flexibility is particularly valuable:

pathologists and technicians can section, stain, and review tissue without immediate pressure to run the spatial assay, which is harder to achieve with platforms like Xenium. Importantly, only the region exposed to probes yields data, so even small misalignments inside the CytAssist can result in profiling empty slide areas. One practical tip is to stain only the hybridized region with eosin (e.g., by pipetting eosin directly into the well without removing the cassette) to produce a visible pink square (~6.5 × 6.5 mm) for accurate alignment before loading into the CytAssist instrument.

Real-time quality control should be embedded throughout (see Box S4 in the supplemental information online). Before starting the experiment, review tissue integrity on a consecutive section using H&E or DAPI staining. Assess staining uniformity and morphology under the microscope to determine whether the sample is suitable for spatial processing. During library prep, measure fragment size and yield.

When things go wrong, act quickly. Even with careful planning, spatial experiments can fail, typically during tissue detachment, hybridization, or transfer steps. Maintain detailed records of all reagents, steps, and images to support troubleshooting. See Box 1 for common failure points and response strategies.

Box 1. When things go wrong: common failures and how to respond

Even well-planned spatial transcriptomics experiments can fail. Understanding where issues commonly arise and how to respond can help recover data, avoid repeating errors, and facilitate troubleshooting with reagent providers.

(i) Document every step

Thorough documentation is essential. Photograph critical steps (e.g., H&E-stained slides, tissue placement on capture areas, hybridization setup), retain all reagent labels and lot numbers, and use a standardized checklist/standard operating procedures to track protocol execution. These records are often required for troubleshooting with technical support and are invaluable when reviewing what went wrong.

(ii) Common points of failure

H&E staining (FFPE samples). A frequent issue occurs during deparaffinization, where tissue detaches from the slide. Using coated slides (e.g., Superfrost Plus) and careful handling can help, but some tissue types remain prone to detachment. Avoid prolonged water baths and monitor section integrity closely during this step.

Probe hybridization (overnight step). Tissue detachment can also occur during hybridization. Once tissue lifts, the experiment cannot be recovered.

(iii) Contact technical support immediately

If an issue occurs, contact the platform provider's support team as soon as possible. Include complete documentation: photos, protocol steps, reagent batch numbers, and RNA quality metrics. Avoid continuing the experiment or discarding materials before receiving guidance, as this may limit troubleshooting options.

(iv) Adhere strictly to the vendor-provided protocol

Reagent providers are best equipped to assist when the standard protocol has been followed exactly. Even small deviations, such as altered incubation times, reagent substitutions, or changes in temperature, can impact performance and reduce the likelihood of successful recovery.

(v) Visium HD users: pay special attention to probe release mix addition

In the Visium HD CytAssist workflow, the step of adding the probe release mix to the Visium HD slide is critical. A common issue occurs when the mix does not evenly spread across the entire capture area. We and others have observed this by inspecting eosin-stained slides after CytAssist transfer, where incomplete staining of the capture area correlates with a lack of transcriptomic signal in the corresponding regions. Ensure that the solution stays exactly at the designated region. Errors at this stage can severely compromise data quality, even if the tissue appears morphologically intact.

Finally, in high-throughput or multi-sample studies, samples should be randomized and processed in parallel wherever possible. Standardized operating procedures, executed by the same technician, help reduce technical variability and batch effects.

ST workflows demand rigor, but they reward attention to detail. Strong execution, rooted in careful preparation, visual inspection, and proactive troubleshooting, is what transforms a well-designed experiment into interpretable, high-quality spatial data.

Step 7. Sequencing decisions for ST libraries

Sequencing is one of the final steps in ST. While often treated as a checklist item, sequencing depth has a major impact on downstream resolution, sensitivity, and interpretability.

For sequencing-based platforms like Visium and Visium HD, manufacturer guidelines often recommend targeting 25 000–50 000 reads per spot. However, in our experience, this is rarely sufficient for FFPE samples or complex tissues. Libraries sequenced at 25 000 reads per spot often fail to recover sufficient transcript complexity (Figure 2F). We routinely aim for 100 000–120 000 reads per spot in FFPE Visium experiments and even higher for Visium HD, where increased resolution demands deeper coverage to distinguish subtle patterns and low-abundance transcripts.

It is important to interpret saturation metrics with caution. Sequencing saturation reflects the proportion of reads that contribute new UMIs, but high saturation does not always indicate high data quality. Libraries derived from low-quality or degraded RNA (low DV200) often saturate prematurely because they contain fewer distinct molecules to begin with. In these cases, saturation values may be misleadingly high, despite shallow transcript diversity and poor gene coverage. More informative metrics include total UMIs, the relationship between read depth and yield, and spatial consistency across spots. While there is no clear correlation between reads per spot and median genes per spot (Figure 2G), this is due to the interplay of RNA quality and tissue complexity. Samples with low DV200 may saturate quickly but yield poor diversity, while high-quality or complex samples may benefit from deeper sequencing without saturating. Therefore, we prioritize deeper coverage (~100 000–120 000 reads per spot for FFPE) to compensate for these factors and improve sensitivity to low-abundance or spatially restricted transcripts (Figure 2H). Budget constraints are real, but sequencing fewer samples well is almost always better than sequencing many poorly. Inadequate read depth often wastes tissue, time, and sequencing resources by generating data that cannot support robust analysis.

Finally, multiplexing samples per lane should be done with caution. Tissue area and RNA quality vary between samples, affecting read distribution. To avoid imbalances, normalize based on tissue coverage and RNA integrity when pooling, and avoid combining poor- and high-quality libraries in the same lane. Ultimately, sequencing is not just a technical endpoint, but also a strategic decision. Planning for sufficient read depth from the outset ensures that spatial data is interpretable and actionable.

Step 8. Data processing, normalization, and interpretation

ST data are not just large, they are also complex. Unlike bulk or single-cell RNA-seq, spatial data layers molecular profiles onto physical coordinates, introducing unique opportunities for biological insight but also new demands for computational care. Equally important is preparing the necessary computational infrastructure. ST datasets can reach hundreds of gigabytes, and underestimating storage and processing needs is a common oversight.

The first step is quality control. Whether spot-based (e.g., Visium) or cell-based (e.g., Xenium), filtering low-quality spots or cells is essential. Remove entries with very low transcript counts, excessive mitochondrial content (if applicable), or other platform-specific outliers [38,39]. Normalization corrects for technical variation in sequencing depth or signal intensity. For spot-based platforms, methods like SCTransform or log normalization are widely used [40]. This step is essential for making transcript counts comparable across spatial units. Next is dimensionality reduction and clustering, which allow researchers to identify spatial domains, cell types, or tissue compartments. In spot-based platforms, this often includes deconvolution to infer underlying cell types using tools like RCTD [41] or cell2location [42] in combination with matched scRNA-seq references. For imaging-based platforms, segmentation and cell typing precede higher-order modeling of cell neighborhoods or gradients.

Batch effects are common in multi-sample studies, especially when RNA quality, tissue area, or platform batches vary. While tools like Harmony, Seurat, or LIGER can correct batch effects, caution is advised: overcorrection can obscure real biological differences. Ideally, experimental design (Step 3) should minimize batch effects at the source.

Spatial data also violates a core assumption of classical statistics: independence between observations. Nearby cells or spots often share similar expression profiles due to tissue architecture or shared microenvironments. Ignoring this spatial autocorrelation can inflate false positives or obscure genuine structure. Metrics like Moran's I or Geary's C help quantify spatial structure and support more robust modeling.

Teams should also anticipate the storage and data management needs of spatial projects. Even without raw imaging files, processed outputs from platforms like Xenium or CosMx can exceed several dozen gigabytes per sample, and cohort-scale studies often require hundreds of gigabytes or more. We recommend adopting consistent file naming, structured metadata documentation, and regular backups from the outset. For data sharing, repositories such as GEO, EGA, and emerging spatial-specific platforms (e.g., SpatialDataHub) support deposition of spatial matrices, coordinate files, and accompanying metadata in standardized formats, facilitating reproducibility and reuse.

Finally, spatial analysis gains power through integration. Combining ST with matched scRNA-seq, clinical metadata, or proteomics enhances interpretability and opens the door to predictive modeling. Cell type mapping, trajectory inference, and neighborhood analysis can reveal how local architecture shapes biological function or therapeutic response. We recommend users to follow on community efforts to identify the most robust tools and best practices [43–50].

Concluding remarks and future perspectives

ST is rapidly evolving from a discovery tool into a core technology for translational research. Advances in resolution, panel design, and throughput are enabling more precise mapping of cell–cell interactions, tumor architecture, and microenvironmental cues across tissue types and disease states, even in 3D [51–53].

One of the most promising frontiers is the integration of spatial with multi-omics. Combining transcriptomics with proteomics, epigenomics, or metabolomics allows for richer models of tissue biology. Multiplexed platforms already support RNA and protein co-detection, and workflows linking spatial data to mass spectrometry or imaging-based proteomics are beginning to mature. These integrations help resolve questions that transcriptomics alone cannot answer.

At the same time, artificial intelligence (AI) is redefining how we extract meaning from spatial data [54–56]. Deep learning models and graph-based neural networks can enhance cell type

Outstanding questions

How can spatial transcriptomics workflows be standardized across platforms and laboratories to enable true cross-study comparability?

What are the minimal sample quality, capture area, and gene panel requirements for reliable detection of clinically relevant spatial biomarkers?

How can sequencing depth and panel design be optimized for different tissue types and preservation methods without compromising sensitivity?

What analytical frameworks are best suited to integrate spatial transcriptomics with other modalities, such as spatial proteomics or multiplex imaging, in a clinical setting?

How can we accelerate the translation of spatial transcriptomics from retrospective studies into prospective clinical trials?

classification, infer tissue domains, and model spatial dependencies that elude traditional clustering. Importantly, AI also enables predictive applications linking spatial phenotypes to treatment response or prognosis.

Looking ahead, the clinical translation of spatial technologies will depend on more than resolution or gene count. It will require scalable, interpretable workflows that can integrate with pathology pipelines. In oncology, spatial biomarkers are already being explored for patient stratification, but broader adoption will hinge on the development of robust pipelines that support regulatory validation [57].

Finally, the field must also grapple with scale. Small-sample studies remain valuable for mechanistic insight, but understanding patient variability will require cohort-scale spatial analysis. We believe this shift is both necessary and feasible provided the community embraces standardization and metadata harmonization.

ST has reached a turning point. What comes next will be defined not only by technological innovation, but by the ability to deploy spatial biology systematically, reproducibly, and in direct service of translational goals (see Outstanding questions).

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Declaration of generative artificial intelligence and artificial intelligence-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT in order to edit the text. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of interests

The authors have no interests to declare.

Supplemental information

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